Report: Performing Gaussian Mixture Model (GMM) on W40 SOFIA CII data cube

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1 Introduction

CII[158μm] line often shows non-single-gaussian, complex spectral profile. Rather than analysing the spectra one by one, It would be beneficial to cluster spectra with similar profile together.

Several papers have utilised Gaussian Mixture Model (GMM) to do the clustering (for example: Kabanovic et al. 2022; Bonne et al. 2023; Tiwari et al. 2023). They will be the main references to this exercise.

2 Data

The target of this exercise is the W40 star forming region. Nicola has kindly provided a PCA-reduced SOFIA-observed CII data cube. The cube has also beed smoothed to 20" spatially and 0.3 km/s spectrally, and its high-noise edges are cropped. The cube has a size of 21.5'$\times$21.5', centered around 18h21m25s -2d3m52.5s. The cube has non-uniform noise with a mean rms noise of 0.75 K T(mb), and the signal-to-noise ratio (SNR) goes up to $\approx$ 100.

3 Data availability

The code is available in the GitHub repository `https://github.com/Lim1029/W40-CII-Analysis`

References

Bonne, L. et al. (Nov. 1, 2023). "The SOFIA FEEDBACK [CII] Legacy Survey: Rapid Molecular Cloud Dispersal in RCW 79". In: *Astronomy & Astrophysics* 679, p. L5. DOI: 10.1051/0004-6361/202347721.

Kabanovic, S. et al. (Mar. 1, 2022). "Self-Absorption in [C II], 12CO, and H I in RCW120 - Building up a Geometrical and Physical Model of the Region," in: *Astronomy & Astrophysics* 659, A36. DOI: 10.1051/0004-6361/202142575.

Tiwari, M. et al. (Dec. 1, 2023). "Identifying Physical Structures in Our Galaxy with Gaussian Mixture Models: An Unsupervised Machine Learning Technique". In: *The Astrophysical Journal* 958.2, p. 136. DOI: 10.3847/1538-4357/ad003c.